

We are using AI whether we like it or not. We ask questions or ask it to perform tasks. AI uses a tremendous amount of electricity. Will peak demand for electricity shift?

Posted by u/Stock_Pen_4019 • 0 points • 4 comments

The companies are generating their own electricity in response to this. But, since the use of AI will match our awake times, will the peak demand for electricity increase due to AI? Can some AI activity be shifted to periods when electrical demand is normally low? Will we be able to accept the answer tomorrow instead of just now?

Comments

u/Majromax • 3 points • 2025-10-03 13:53 UTC

> But, since the use of AI will match our awake times, will the peak demand for electricity increase due to AI?

"The" peak is a bit of a misnomer. Current datacenter energy use seems to be more of a strain on local infrastructure, particularly distribution, rather than national infrastructure.

Remember that AI is far from the only energy-intensive industry. Aluminum smelting, for example, is famously power hungry. The main difference is that AI is new and that it can expand quickly, meaning hungry providers are more likely to be more nimble than old and slow-moving infrastructure plans and regulations. If a region is offering cheaper power and connections than it should, a dozen data centres can spring up before regulators have time to blink.

> Can some AI activity be shifted to periods when electrical demand is normally low? Will we be able to accept the answer tomorrow instead of just now?

This kind of shift is possible, but remember that power is only one part of the AI equation; the other is expensive computer hardware. A GPU that's sitting unpowered provides no value.

For a rough comparison, an NVidia [H100]([https://en.wikipedia.org/wiki/Hopper_\(microarchitecture\)](https://en.wikipedia.org/wiki/Hopper_(microarchitecture))) takes about 700 watts of power, and looking it up on eBay gives a cost of about \$35,000 per card. Assuming the card depreciates over three years^{†}, owning it has a "cost" of about \$12,000 per year, or \$1.33 per hour.

In contrast, the [average power cost](<https://www.electricchoice.com/electricity-prices-by-state/>) for commercial users appears to be about 25¢ per kWh, implying that running the H100 at full capacity

costs \$0.18 per hour in electricity.

Idling the GPU to save on power throws away dollars to save dimes, even before we consider the opportunity cost of delaying the presumably-valuable work being performed.

^(† — This is a bit aggressive, but in practice it won't be straight-line depreciation to zero but more of an exponential decay. I'm considering the current rate of depreciation rather than the long-term residual value.)

u/Hivemind_alpha • 3 points • 2025-10-03 12:16 UTC

Why else do you think the backers of AI companies are also backing fusion startups?

u/Stock_Pen_4019 • 1 points • 2025-10-03 12:18 UTC

They are forward thinking risk takers. I on the other hand have been around for a long time, almost as long as smart people have been promising, cheap electricity from fusion power.

u/Putnam3145 • 2 points • 2025-10-03 13:43 UTC

With essentially no money actually going into it, yes.